

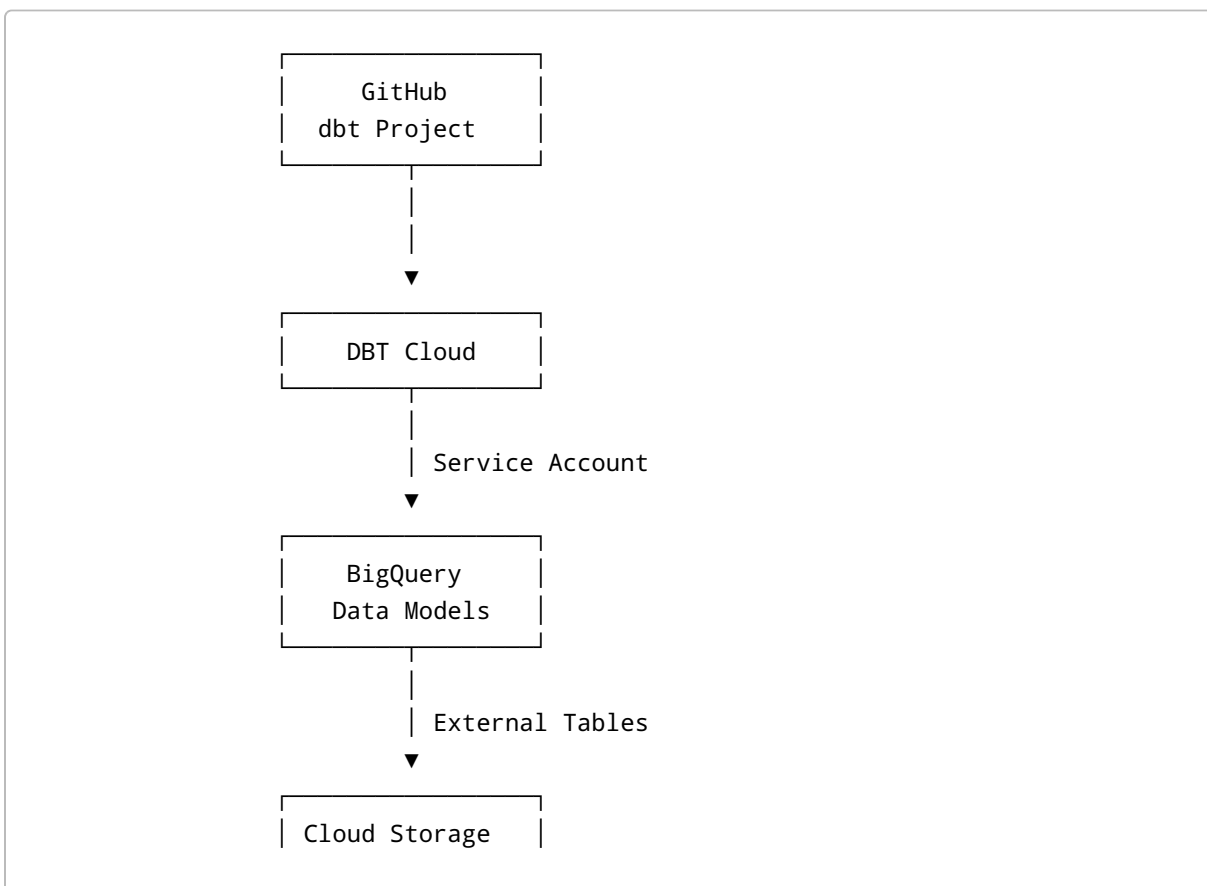
Creating a Google BigQuery Project and DBT Cloud Environment for an End-to-End Analytics Project

Overview

This guide explains how to:

1. Create a Google Cloud Project
2. Create a Service Account
3. Generate Service Account Keys
4. Create a Cloud Storage Bucket
5. Upload Mock Data
6. Create BigQuery External Tables
7. Configure a DBT Cloud Project
8. Connect DBT Cloud to GitHub

Architecture Overview



Prerequisites

Before starting, ensure you have:

- A Google Account
- Access to Google Cloud Platform (GCP)
- A GitHub Account
- A DBT Cloud Account

Useful services:

- Google Cloud Console
- BigQuery
- Cloud Storage
- DBT Cloud
- GitHub

Step 1: Create a Google Cloud Project

1. Open Google Cloud Console.
2. Click the Project Selector at the top of the page.
3. Click **New Project**.
4. Complete the project information:

Field	Example
Project Name	dbt-bike-share-project
Organization	Your organization OR No Organization

Important

If this is a personal learning project:

- Select **No Organization**

If this project belongs to a company:

- Select the appropriate organization.

Some organizations restrict the creation of Service Account keys. Using **No Organization** for personal projects generally avoids these restrictions.

1. Click **Create**.
-

Step 2: Select the New Project

After the project is created:

1. Click the project selector at the top of the page.
2. Select the newly created project.

Verify that the correct project name appears in the top navigation bar.

Step 3: Enable Required APIs

Before creating resources, enable the required services.

Navigate to:

APIs & Services
→ Library

Enable:

- BigQuery API
- Cloud Storage API
- IAM API

Wait until all services are enabled.

Step 4: Create a Service Account

Navigate to:

IAM & Admin
→ Service Accounts

Click:

Create Service Account

Complete the form:

Field	Example
Service Account Name	dbt-service-account
Service Account ID	Auto-generated
Description	Service Account for DBT Project

Click:

Create and Continue

Step 5: Assign Roles

Grant the Service Account permissions.

Add the following roles:

Role	Purpose
BigQuery Admin	Manage BigQuery datasets and tables
Storage Admin	Access Cloud Storage buckets and files

Click:

Continue

Step 6: Grant User Access (Optional)

If other users need access:

Add them under:

Grant users access to this service account

Examples:

- Team members
- Instructors
- Project collaborators

Click:

Done

Step 7: Generate a Service Account Key

After the Service Account is created:

1. Click the Service Account.
2. Open the **Keys** tab.
3. Click:

Add Key
→ Create New Key

1. Select:

JSON

1. Click:

Create

A JSON file will automatically download.

Important

Keep this file safe.

This file will later be used by DBT Cloud to connect to BigQuery.

Example:

```
dbt-service-account.json
```

Do not share this file publicly or upload it to GitHub.

Step 8: Create a Cloud Storage Bucket

Navigate to:

```
Cloud Storage  
→ Buckets
```

Click:

```
Create Bucket
```

Provide:

Field	Example
Bucket Name	dbt-bike-share-data
Region	Choose your preferred region
Storage Class	Standard

Click:

```
Create
```

Step 9: Upload Mock Data

Generate the mock data using the scripts provided in the GitHub repository.

After generating the files:

1. Open the bucket.
2. Click:

Upload Files

1. Upload the CSV files.

You can verify the upload by checking:

Cloud Storage
→ Bucket
→ Uploaded Files

Step 10: Create BigQuery External Tables

External Tables allow BigQuery to query files directly from Cloud Storage.

Benefits:

- No duplicate storage
- Lower costs
- Single source of truth

Architecture:

```
graph TD
    A[CSV Files] --> B[Cloud Storage Bucket]
    B --> C[External Table]
    C --> D[BigQuery Queries]
```

Run the SQL scripts provided in the GitHub repository.

The scripts will:

- Create datasets
- Create external tables
- Connect BigQuery to files stored in Cloud Storage

Verify the tables appear under:

```
BigQuery
  → Dataset
    → External Tables
```

Step 11: Create a DBT Cloud Project

Log in to DBT Cloud.

Click:

Create New Project

Provide:

Field	Example
Project Name	Bike Share Analytics
Warehouse	BigQuery

Step 12: Configure BigQuery Connection

During setup:

1. Choose:

BigQuery

1. Authentication Method:

Service Account JSON

1. Upload the JSON key created earlier.

DBT Cloud will use this Service Account to connect to BigQuery.

Step 13: Connect GitHub Repository

In DBT Cloud:

1. Select:

Connect Repository

1. Authenticate with GitHub.
2. Authorize DBT Cloud.
3. Select the repository containing your DBT project.

Example:

github.com/username/dbt-bike-share-project

1. Click:

Create Project

Step 14: Verify Everything Works

Test the connection from DBT Cloud.

Verify:

✓ BigQuery Connection Successful

✓ Repository Connected

✓ Development Environment Created

✓ Models Visible

You are now ready to:

```
dbt run
dbt test
dbt build
```

and build your end-to-end analytics project.

Final Architecture

